LAB SUMMARY

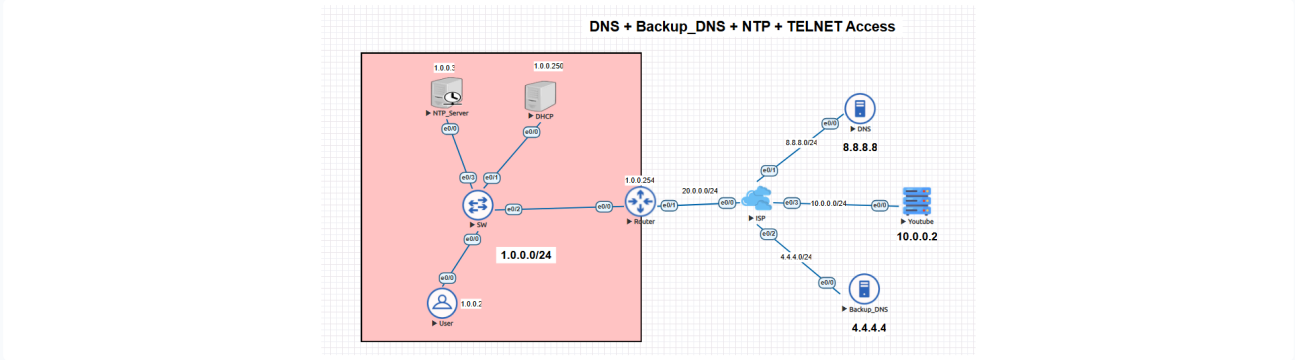
Network Services Implementation: DNS, NTP & Telnet

Concise record of topology setup, tasks performed, and verification results.

1. Topology Design & IP Addressing

Overview

- **Local LAN (1.0.0.0/24):** Configured client host USER (1.0.0.2), local NTP_Server (1.0.0.3), and DHCP (1.0.0.250) interconnected via switch SW.
- **Gateway & WAN:** Default gateway configured on Router (LAN: 1.0.0.254, WAN: 20.0.0.1/24) connecting to the ISP cloud.
- **External Services:** Deployed DNS (8.8.8.8), Backup_DNS (4.4.4.4), and remote server Youtube (10.0.0.2) through the ISP.



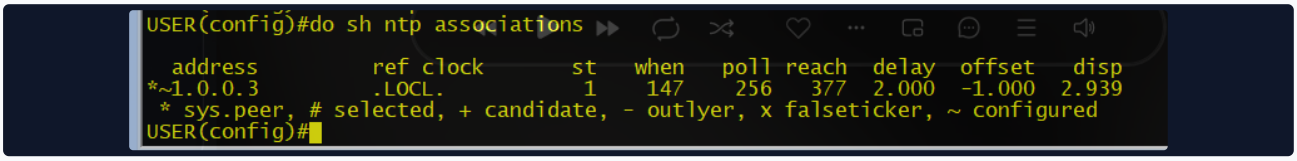
Reference Image 1: Complete Lab Topology Diagram — LAN 1.0.0.0/24 • WAN 20.0.0.0/24

Device Node	Interface	Assigned IP / Mask	Function in Lab
USER	e0/0	1.0.0.2 /24	Client machine testing NTP, DNS lookup & Telnet access
NTP_Server	e0/0	1.0.0.3 /24	Local Stratum 1 reference clock for time sync
Backup_DNS	e0/0	4.4.4.4 /24	Secondary nameserver resolving FQDN to target server
Youtube	e0/0	10.0.0.2 /24	Remote destination host running Telnet service

2. NTP Time Synchronization Setup

Task 1

- **What was done:** Configured USER to synchronize its system time from local NTP server `1.0.0.3`.
- **Verification command:** Executed `do sh ntp associations` on the client terminal.



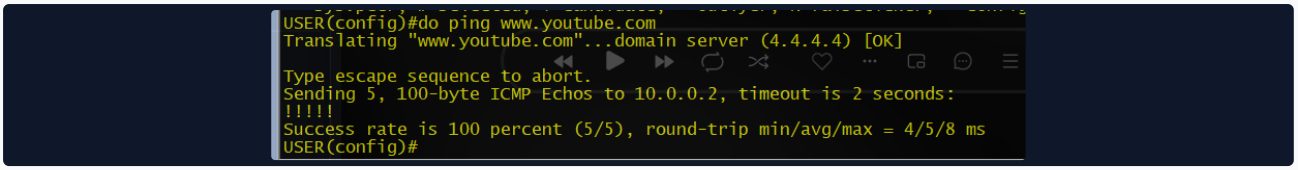
Reference Image 2: NTP Association Terminal Output — Status: Synchronized (sys.peer)

Result: *~`1.0.0.3` confirms the server is statically configured (~) and chosen as active system peer (*) at Stratum 1 (`.LOCL.`) with full reachability (377).

3. DNS Resolution & Ping Reachability

Task 2

- **What was done:** Configured dual DNS servers (Primary `8.8.8.8`, Backup `4.4.4.4`) on client USER and tested resolution for `www.youtube.com`.
- **Verification command:** Executed `do ping www.youtube.com` from `USER(config)#`.



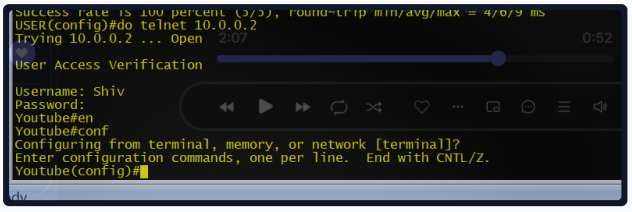
Reference Image 3: Domain Translation via Backup DNS & ICMP Echoes **Success Rate: 100% (5/5)**

Result: Domain translated successfully via **4.4.4.4 [OK]** to **10.0.0.2**. All 5 ICMP echo packets returned successfully with round-trip average of 5 ms.

4. Remote Telnet Access (IP vs. FQDN)

Task 3

- **What was done:** Enabled Telnet (VTY lines) with user `Shiv` on the target server. Tested login from USER using both direct IP address and domain name.



Reference Image 4: Telnet via IP (10.0.0.2)



Reference Image 5: Telnet via FQDN (www.youtube.com)

- **Direct IP Access:** `do telnet 10.0.0.2` opened connection immediately; authenticated user `Shiv`, entered privileged mode `enable`, and accessed configuration mode `Youtube(config)#`.
- **FQDN Access:** `do telnet www.youtube.com` triggered domain translation via server `4.4.4.4`, resolved to `10.0.0.2`, opened the session, authenticated `Shiv`, and reached `Youtube(config)#`.

5. Task Verification Summary

Status Matrix

Task Item	Command Executed	Target Node / Reference	Result
NTP Clock Sync	`do sh ntp associations`	1.0.0.3 (Image 2)	PASS — Peer synchronized, Stratum 1
Backup DNS Failover	`do ping www.youtube.com`	4.4.4.4 → 10.0.0.2 (Image 3)	PASS — Resolved via 4.4.4.4, 5/5 ping
Telnet (by IP)	`do telnet 10.0.0.2`	10.0.0.2 (Image 4)	PASS — Login verified, config access
Telnet (by Name)	`do telnet www.youtube.com`	www.youtube.com (Image 5)	PASS — DNS resolved & logged in